Final Review – B.TECH PROJECT

Graph-MoE: A Topology-Aware Mixture-of-Experts Framework for Drift Adaptive Weather Forecasting

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Problem statement and motivation

- Weather and climate regimes shift over time (El Niño vs. normal years): a model trained once quietly goes stale as conditions change.
- Retraining the entire model, from scratch or on a fixed schedule, is wasteful: a shift is usually confined to particular regions or relationships.
- Most forecasters choose which regions to trust by correlation or geography, so they cannot tell "this region's physics changed" from "a coincidental correlation broke".
- As systems grow from regional to global scale, the number of regions that can drift grows too, and full retraining becomes even more costly.
- **Our approach:** validate real causal drivers first, then adapt only the regions that actually drifted.

Phase 1 objectives

1. **Causal graph:** discover which regions' past conditions carry information about another region's future OLR (PCMCI over 50 regions x 6 variables).

2. **Stress tests:** check every link against fake decoy variables, physical travel speeds, trimming to the strongest links, and redrawing the regions at double resolution.

3. **Graph as guidance:** feed the validated graph to the causal splitter three ways: narrow its candidates, nudge its scores, or penalise selections outside the graph.

4. **Unseen years:** compare the graph-guided splitter with an unguided one on held-out 2013–2022.

5. **Drift-adaptive experts:** detect drift per region, then retrain, spawn, or reactivate only that region's expert.

Phase 2 objectives

1. **India forecasting backbone:** shared graph neural network on hourly surface data, with region-specific experts conditioned on Phase 1 validated causal drivers.

2. **72-hour forecasts:** autoregressive roll-out to 72 h, with a station-level component checked against real observations.

3. **Benchmark:** compare with GraphCast and WeatherNext on the same data, horizons and evaluation protocol.

4. **Global scale:** extend the validated regional architecture to larger meshes, more regimes and a larger expert pool.

03 | Consolidated Literature Review & Research Gap

Paper	Key idea / method	Limitation
Wu et al., "Discovering invariant rationales for graph neural networks", ICLR 2022	Splits a graph into causal rationale and non-causal part; swaps the non-causal part across instances; stable prediction = invariant causal information.	Built for graph classification; no spatiotemporal forecasting; no dynamic expert adaptation.
Ghaffari et al., "Graph-conditioned mixture of GNN experts for traffic forecasting", arXiv 2026	A small router blends frozen, pre-trained experts per node using topology and recent input.	Experts frozen forever: no retraining, no spawning, no answer to drift.
Cao et al., "Geometric mixture-of-experts with curvature-guided adaptive routing", arXiv 2026	Routes each node to expert sub-networks by Ollivier–Ricci curvature across Riemannian spaces.	Routes by topology/curvature alone: no notion of drift or causality.
Wu et al., "GraphMETRO: mitigating complex graph distribution shifts via mixture of aligned experts", NeurIPS 2024	Each expert is invariant to one hand-constructed shift type; blended by a learned gate.	Shift components hand-constructed and static; no shift structure learned from data.

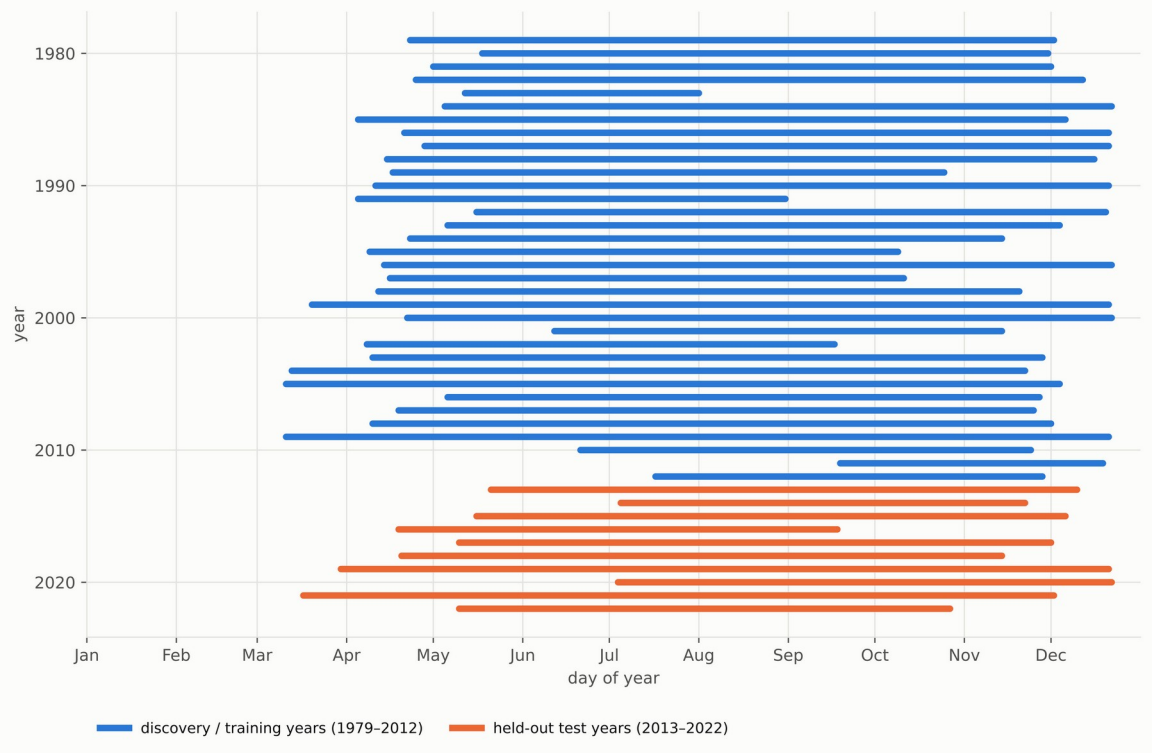
RESEARCH GAP

Existing studies address causal rationale discovery, graph distribution shifts, dynamic expert learning and graph-based weather forecasting separately. None combines causal structure discovery with drift-triggered expert adaptation for spatiotemporal weather forecasting.

06 | Dataset

- **Source:** BSISO monitoring data; the Boreal Summer Intraseasonal Oscillation is a slow-moving pattern of tropical convection related to the MJO.
- **Grid:** 2.5° tropical belt, 30°S–30°N, all longitudes (25 x 144 = 3,600 cells, about 275 km each).
- **Time:** daily, 1979-01-01 to 2022-12-31; 16,071 days, none missing.
- **Fields:** SST, h850, u200, PW, u850, OLR, as anomalies normalised per grid cell.
- **Target:** OLR, a proxy for convection (low OLR = active rain) and the least persistent field.
- **Regions:** 3,600 cells -> 50 regions by k-means on OLR -> 50 x 6 = 300 nodes; the adjacency mesh defines candidate pools.
- **Split:** 1979–2012 for graph discovery and training; 2013–2022 held out for testing only.
- **Season:** Phase 1 scores BSISO-active target days only (208 days/yr on average); source history is kept in full.

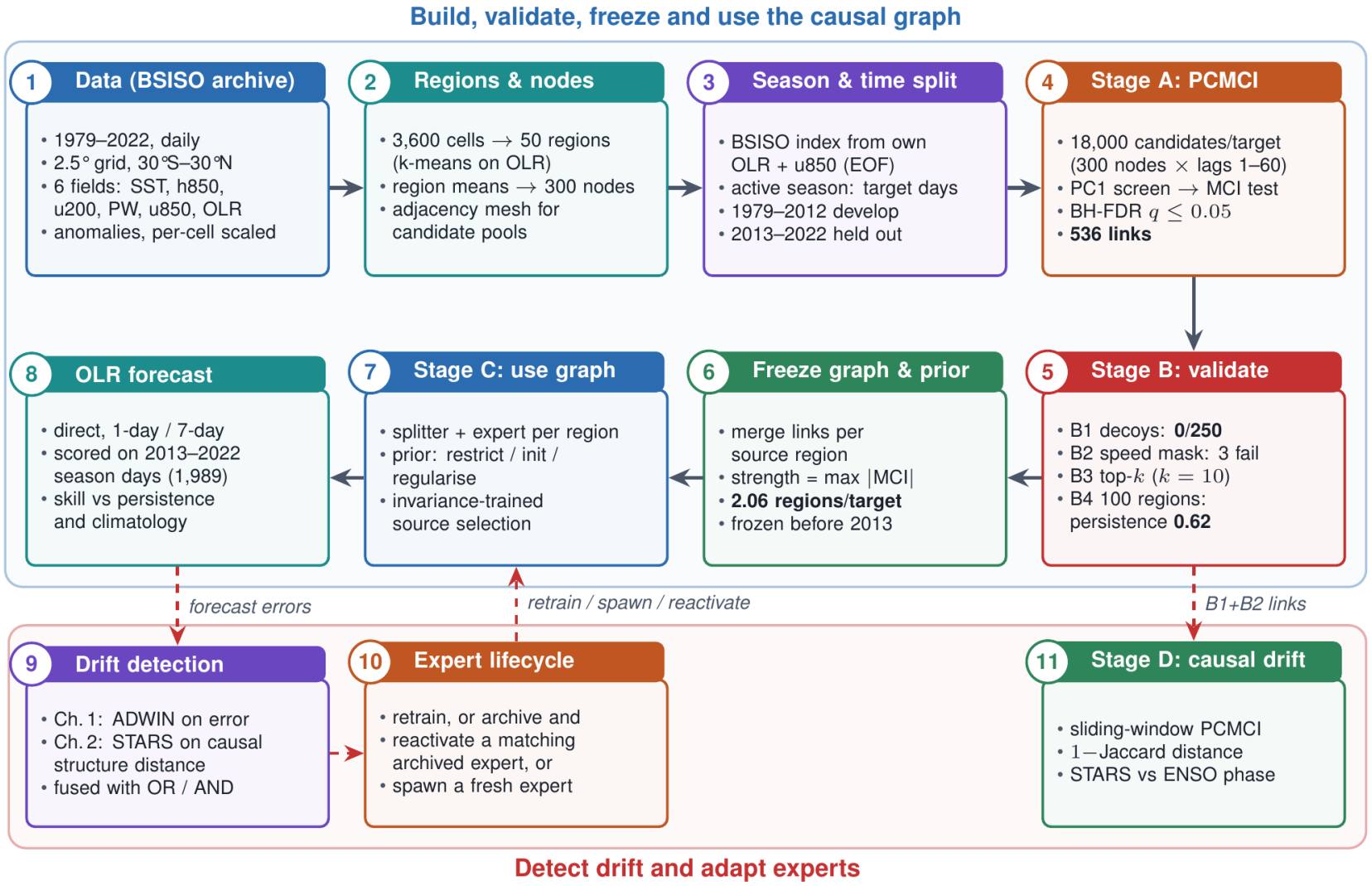
06 | Dataset: active BSISO season



- BSISO index rebuilt from our own OLR + u850 (combined EOF, May–Oct, 1979–2012 only).
- **EOF variance 11.2% / 7.7%** vs. about 10% / 7% published by Lee et al. (2013).
- Season = 31-day mean amplitude > 1 for 20+ days, padded by 21 days.
- **208 days/yr** on average (82–286); onset April–July.

Active season per year: blue = discovery/training years, orange = held-out test years.

System architecture



How PCMCI works?

1. **Build candidates:** For each OLR target, every node (region x field) at every lag 1–60 days: 18,000 candidates per target.

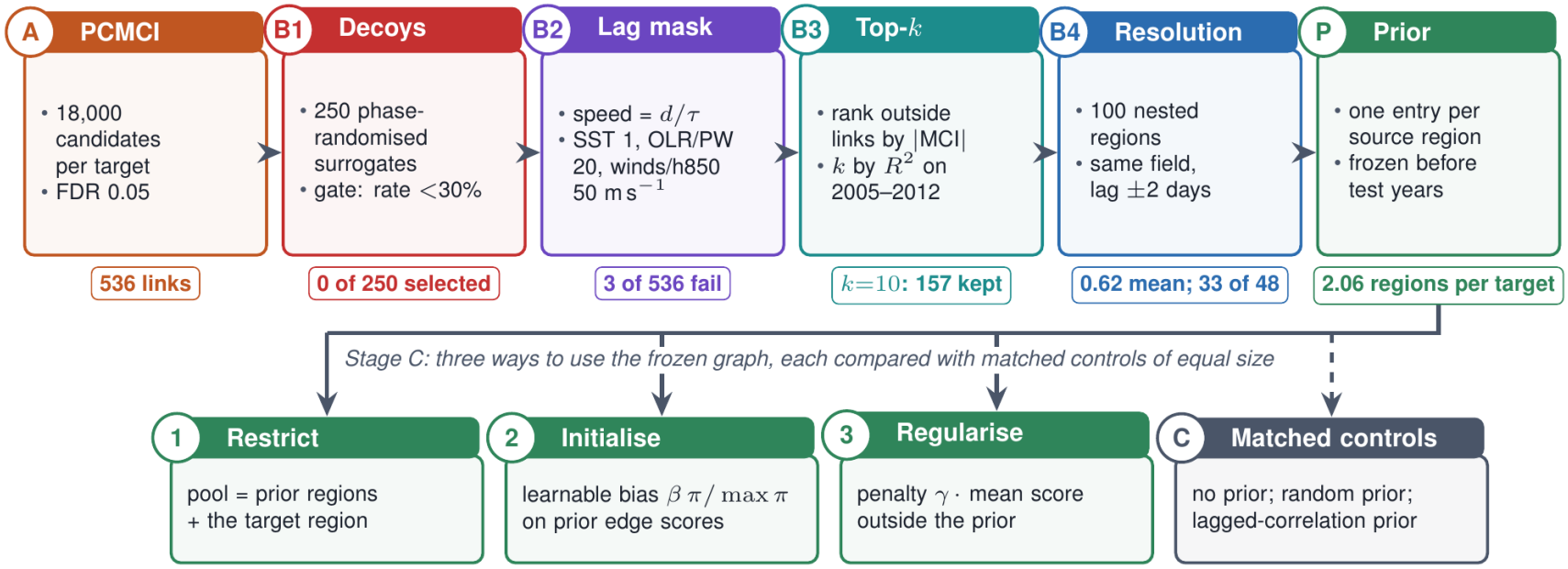
2. **PC1 screening:** Test each candidate against the target while controlling for the strongest candidates found so far; weak candidates drop out early.

3. **MCI final test:** Retest each survivor while controlling for the target's other parents AND the source's own past; a link survives only if it still adds information.

4. **FDR correction:** Benjamini–Hochberg across all 18,000 tests per target keeps only reliable links ($q \leq 0.05$).

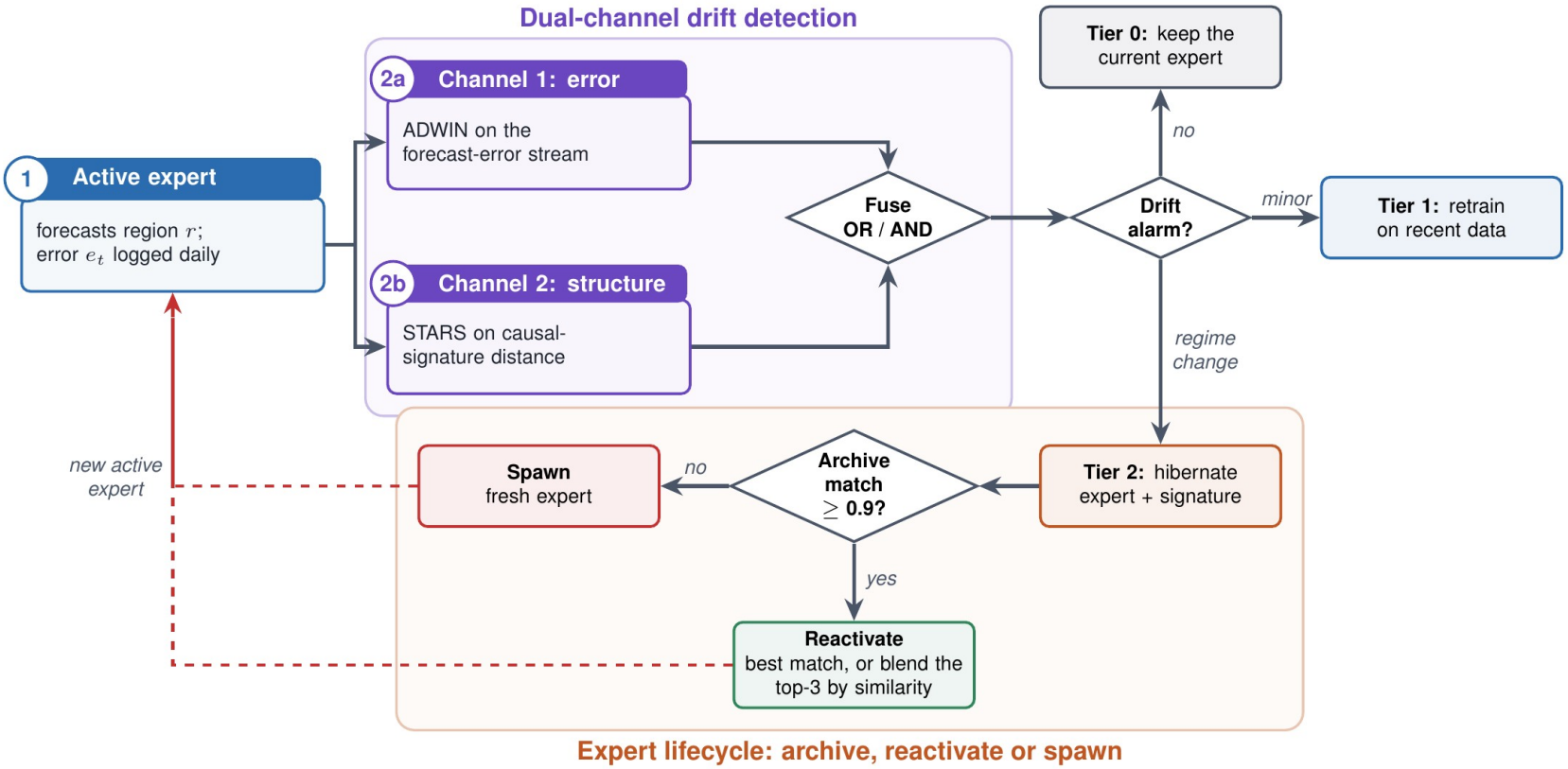
Lag 0 is excluded, so direction comes from time order. Our vectorised implementation matches the Tigramite reference (identical PC1 parent sets, MCI values within 0.002) and runs 50 targets x 18,000 candidates in about 23 minutes.

Graph validation: four independent checks



- **B1 Decoys:** phase-randomised copies keep each series' memory but no real link; does PCMCI pick them?
- **B2 Lag mask:** implied speed = distance / lag must be physically possible for the source field.
- **B3 Top-k:** keep the k strongest outside links per target, k chosen on 2005–2012.
- **B4 Resolution:** do links reappear when every region is split in two?

How the drift detectors and expert lifecycle work



- **Channel 1, ADWIN:** raises an alarm when recent forecast errors differ significantly from older ones.
- **Channel 2, STARS:** tests the causal-signature distance for a confirmed regime shift; it can react before errors rise.
- **Lifecycle:** retrain, or archive and reactivate a matching expert (cosine similarity ≥ 0.9), else spawn a new one.

Model, configuration and parameters

Hyperparameter (splitter + expert)	Value
Epochs / training samples	5 / 4,000 contiguous
Optimiser, learning rate, batch	Adam, 1e-3, 16
Hidden width / generator window	16 / 10 days
Interventions; variance weight	6 swaps; 0 -> 1 after 30% warm-up

Causal graph (PCMCI)	Value
Nodes	300 (50 regions x 6 fields)
Candidates per target	18,000
Lags tested	1–60 days
PC alpha / FDR alpha	0.05 / 0.05
Discovery days	7,153 season days, 1979–2012

- **50 independent splitter + expert pairs:** one per region, trained from scratch, no weight sharing.
- **Splitter (DIR-style):** rationale generator scores each candidate source from 10-day windows; top fraction = causal set; non-causal sources swapped with other days; loss = mean + variance of risk across swaps.
- **Expert:** weighted mean of the selected sources' features (weights from the splitter) -> MLP -> OLR forecast.

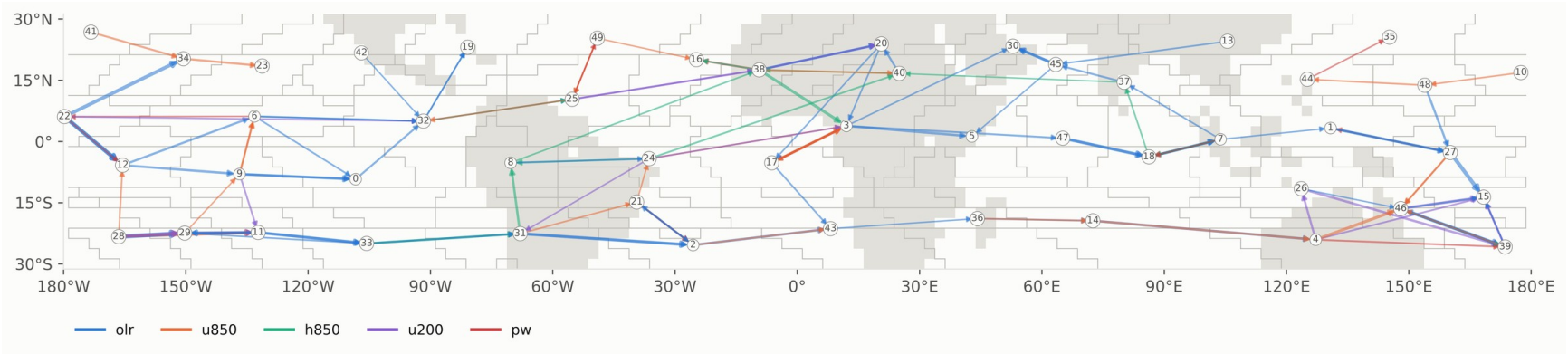
Phase 1 results: the discovered causal graph

536
significant links (FDR 0.05)

160
links from other regions

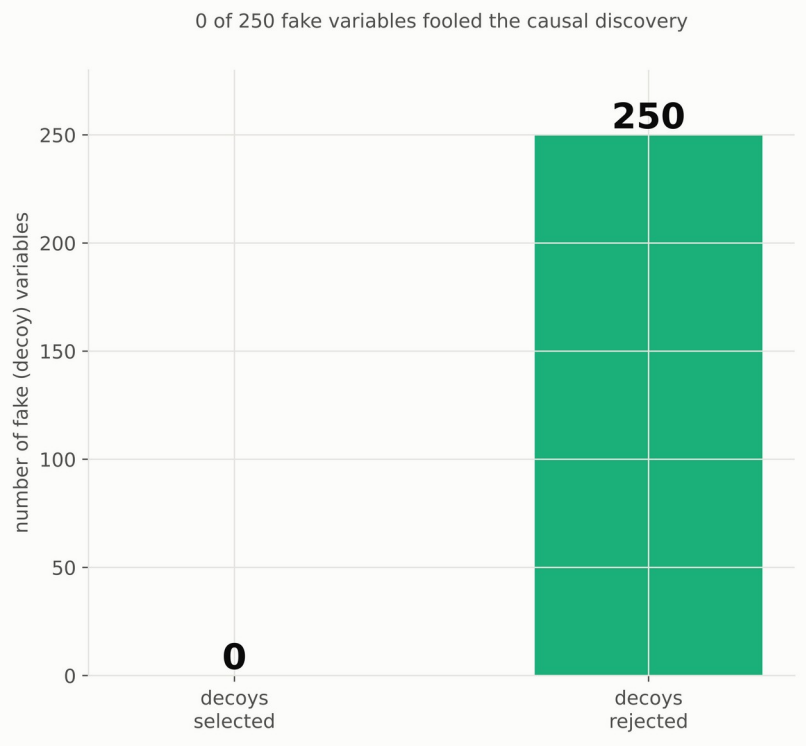
1 day
median lag of outside links

2.06
source regions per target

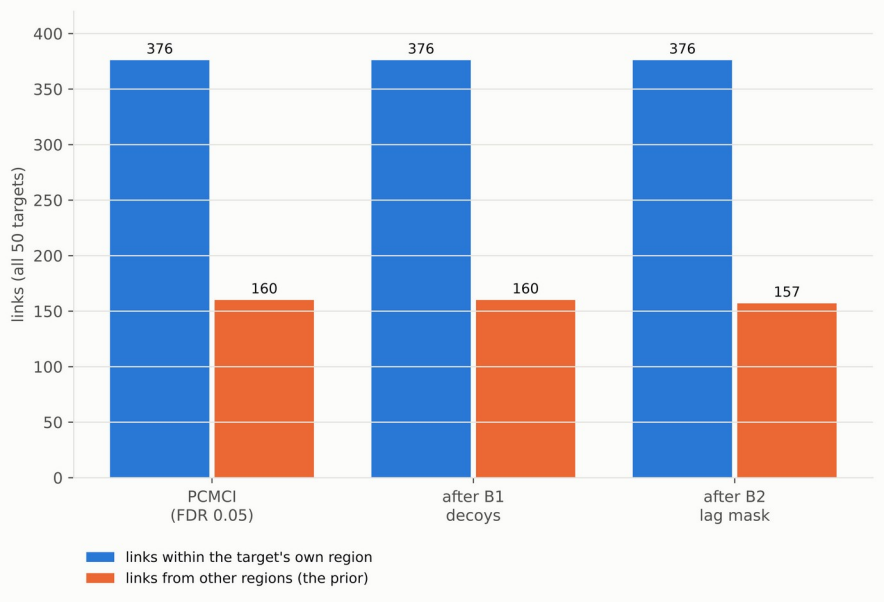


- Outside drivers by field: OLR 72, u850 37, u200 22, h850 18, PW 11, SST 0; mostly neighbouring regions.
- A few long links (wind and height) cross thousands of km in 1–2 days, at equatorial-wave speeds.
- **Region 22:** the graph picks regions 27 and 6; the unguided splitter independently picks 6, 12 and 27. Two different methods agree.

Phase 1 results: B1 decoy test



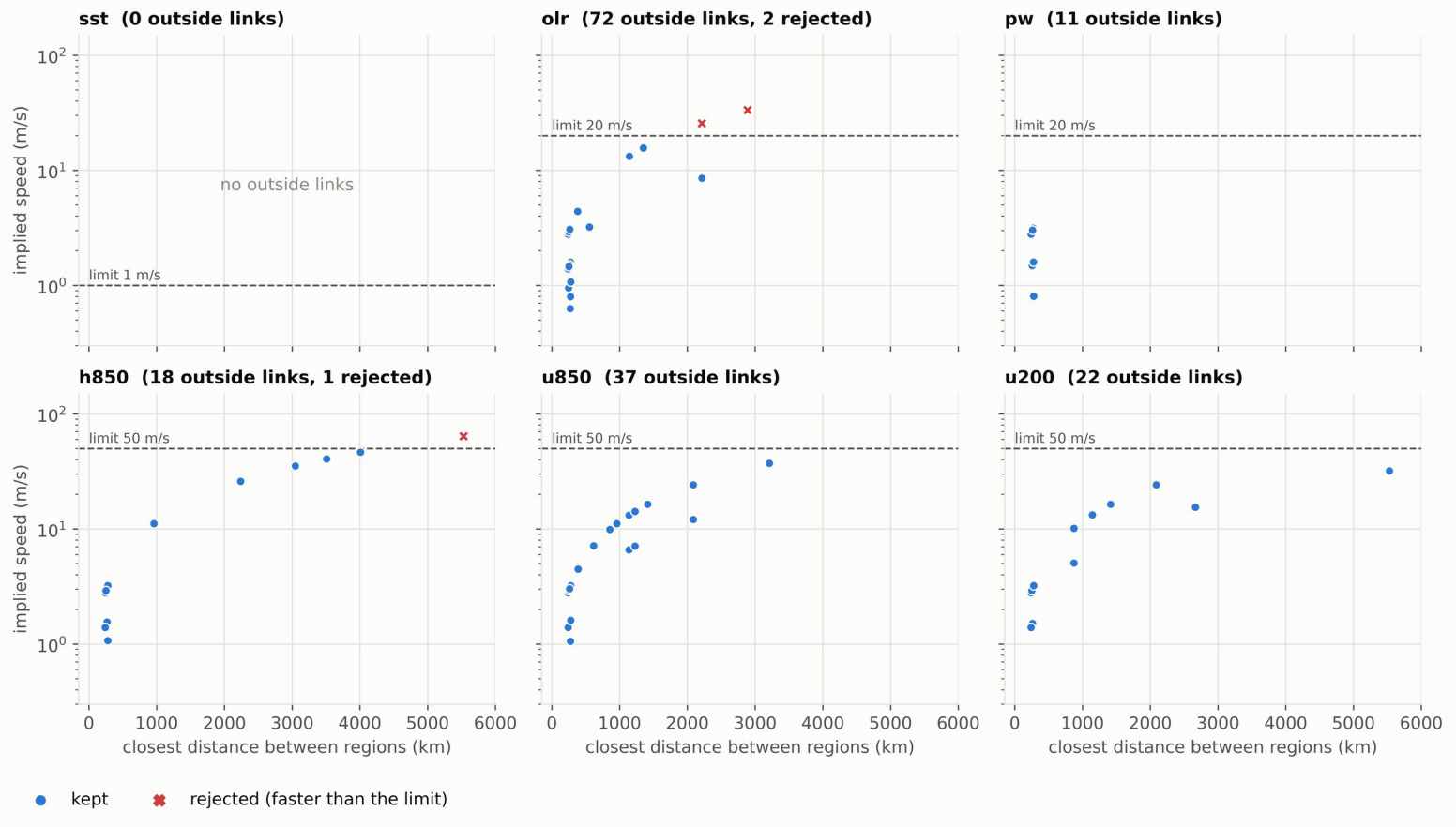
0 of 250 fake variables fooled the causal discovery



Own-region (blue) and outside (orange) links after each check

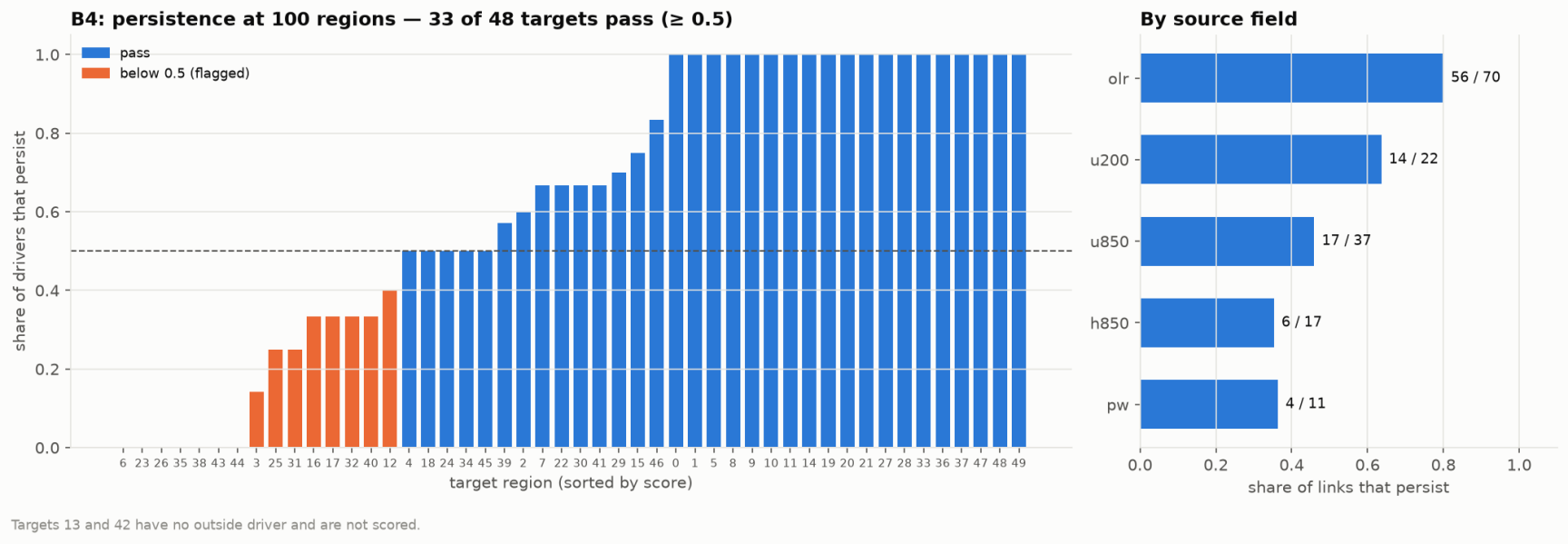
Decoys share the real series' memory and 30–60-day rhythm but have no real link. None was selected (gate: below 30%), and the real graph barely changed (517 vs. 536 links), so PCMCI is not fooled by autocorrelation.

Phase 1 results: B2 physical lag mask



- **Only 3 of 536 links** need an impossible speed (64, 33 and 26 m/s at lag 1); they are dropped from the prior.
- Limits: SST 1 m/s; OLR and PW 20 m/s (coupled Kelvin waves); winds and height 50 m/s.
- 28 kept outside links span 500–5,500 km at up to 46 m/s, all carried by wind and height fields.

Phase 1 results: B3 top-k and B4 resolution



0.62

mean persistence at 100 regions

33 / 48

targets with persistence ≥ 0.5

k = 10

B3 keeps all 157 outside links

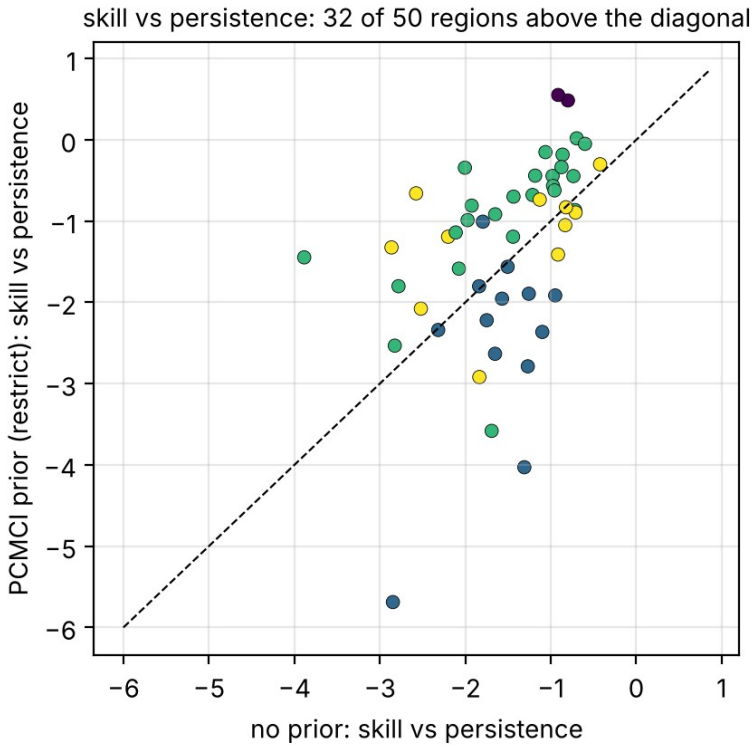
OLR links persist best (56/70); moisture and low-level height links, being more local, least. PCMCI + FDR is already sparse, so B3 keeps every outside link.

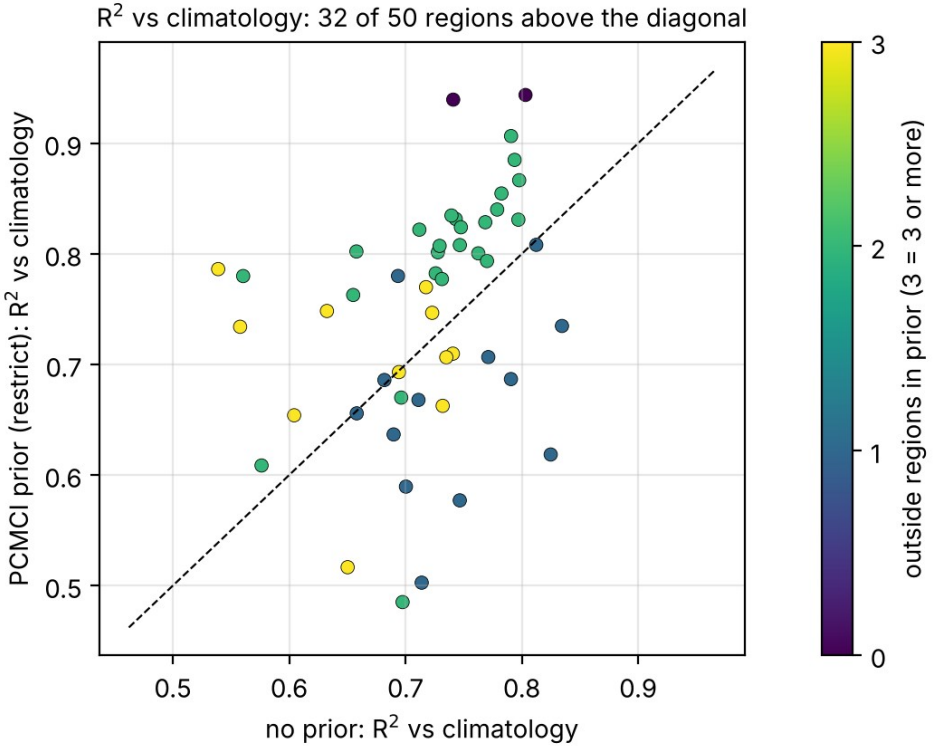
Phase 1 results: causal graph as the splitter's guide

Lead	Prior	Informative splitters	Beat persistence	Mean skill vs pers.	Mean R^2 vs clim.
7 days	none	0/50	50/50	+0.415	-0.015
7 days	PCMCI	**24/50**	50/50	+0.416	-0.015
1 day	none	0/50	0/50	-1.527	+0.720
1 day	PCMCI	**36/50**	3/50	**-1.327**	**+0.745**

- **Full 50-region pool:** without guidance the splitter's scores are uninformative everywhere; with the graph (restrict mode) 24 regions at 7 days and 36 at 1 day become informative.
- **1-day skill improves in 32 of 50 regions** (median +0.40 skill vs persistence, about 17% lower MSE; Wilcoxon p = 0.049).
- **7-day skill unchanged:** the validated links act within 1–2 days, so they help short leads; week-ahead guidance needs a graph built with lags of 7+ days.
- Held-out 2013–2022 season days (1,989); skill = 1 − MSE / MSE(reference).

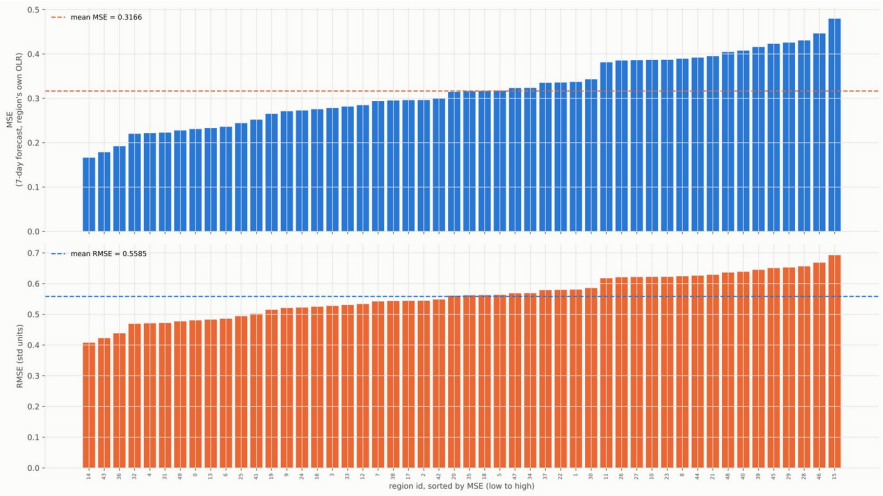
Phase 1 results: 1-day gain, region by region



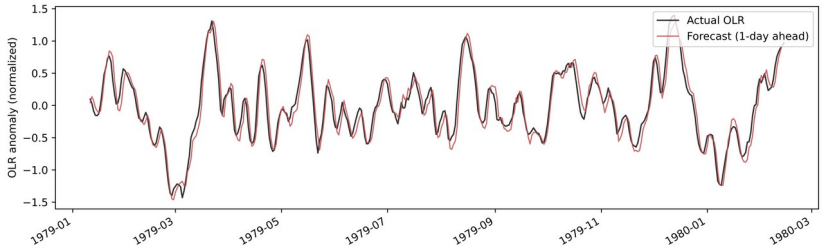


- Each dot is a region; dots above the diagonal improve with the causal prior.
- **Region 22:** skill vs persistence -0.60 -> -0.05.
- Regions with 2+ causal drivers improve on average; regions with a single driver are the hardest case (the splitter must choose between own history and one driver).

Phase 1 results: forecast accuracy



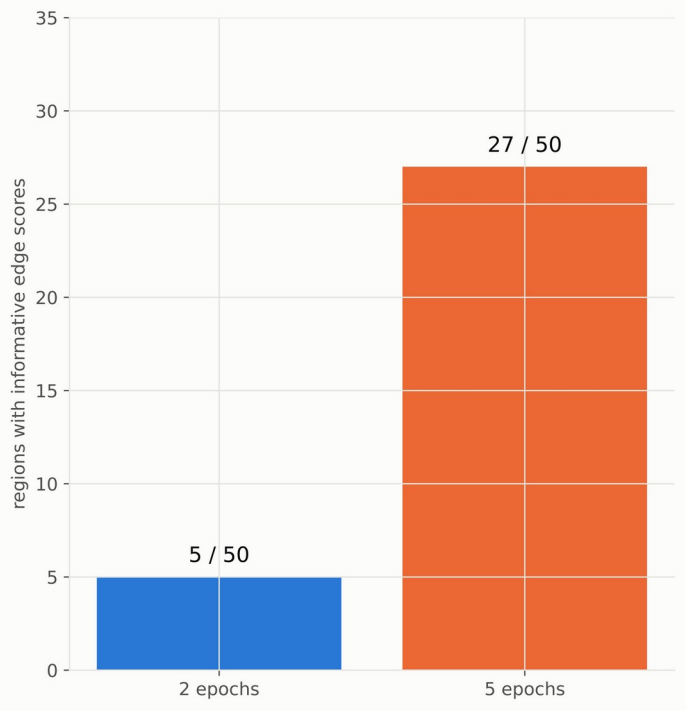
7-day forecast error per region (mean MSE 0.317, RMSE 0.559 std)



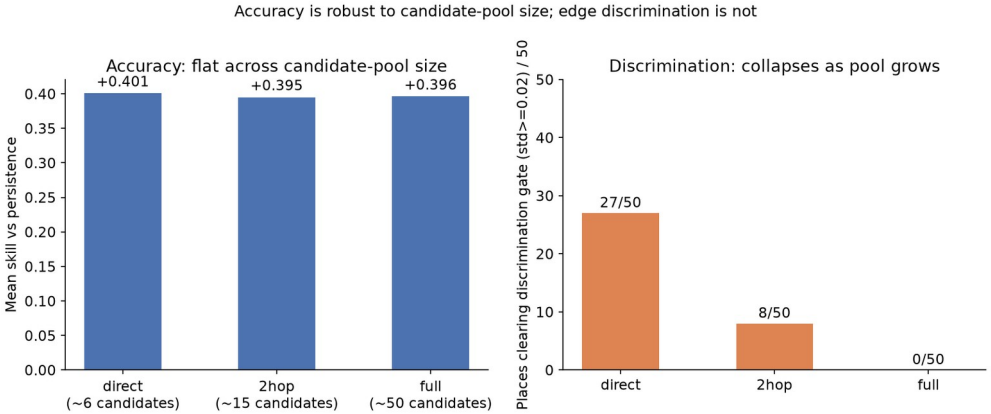
Region 22: 1-day forecast vs observed OLR (illustrative 1979–80 window)

- **Beats persistence in all 50 regions at 7 days** in every candidate pool (mean skill about +0.40, i.e. about 40% lower error than "no change").
- The expert tracks OLR swings one day ahead, including sharp convective transitions.
- At 7 days OLR keeps little memory, so climatology becomes the harder reference (mean R^2 vs climatology about -0.015).

Phase 1 results: what makes source selection work



Training budget: 5/50 -> 27/50 informative



Pool size: accuracy flat, discrimination falls

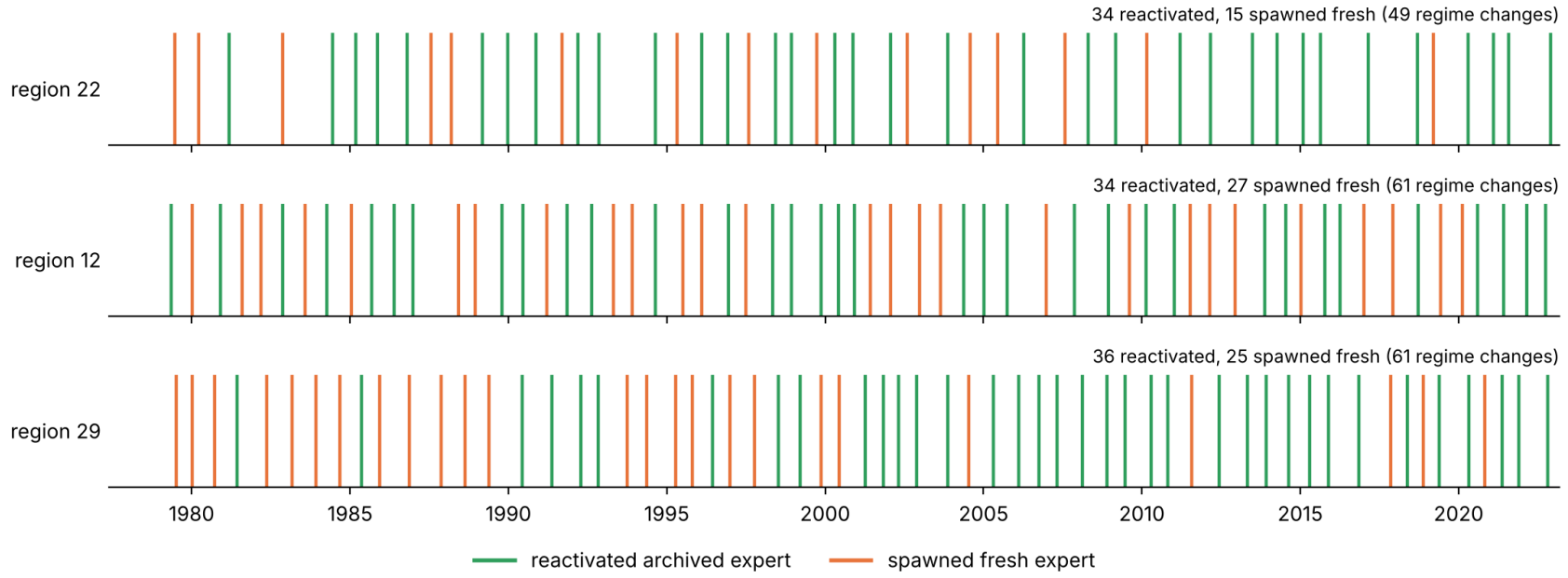
- **Training:** 5 epochs instead of 2 raises informative splitters from 5 to 27 of 50 at equal accuracy.
- **Pool size:** informative splitters fall from 27 (neighbours) to 8 (2-hop) to 0 (all 50 regions). This is exactly the gap the validated graph closes (0 -> 24 / 36).

Phase 1 results: drift detection

Region	Shift	Ch.1 error (ADWIN)	Ch.2 causal (STARS)	Fused OR	Fused AND
22	Abrupt El Niño 97–98	159 d	**35 d**	35 d	missed
22	Gradual 2013–22	966 d	**181 d**	181 d	966 d
29	Abrupt El Niño 97–98	223 d	**92 d**	92 d	235 d
29	Gradual 2013–22	1,158 d	**117 d**	117 d	1,195 d
12	Abrupt El Niño 97–98	**95 d**	123 d	95 d	123 d
12	Gradual 2013–22	**70 d**	100 d	70 d	100 d
False alarms / yr	22 / 29 / 12	0.17 / 0.34 / 1.04	1.62 / 2.25 / 2.19	1.78 / 2.59 / 3.23	0.17 / 0.34 / 0.94

- **The causal channel sees shifts first:** at regions 22 and 29 it detects the El Niño 2.4–4.5x sooner and the gradual drift 5.3–9.9x sooner than the error channel.
- Fused OR catches every shift at every region; AND is slower but raises far fewer false alarms: a real, measured trade-off.
- Latency = days from the start of the shift window to the first alarm in it.

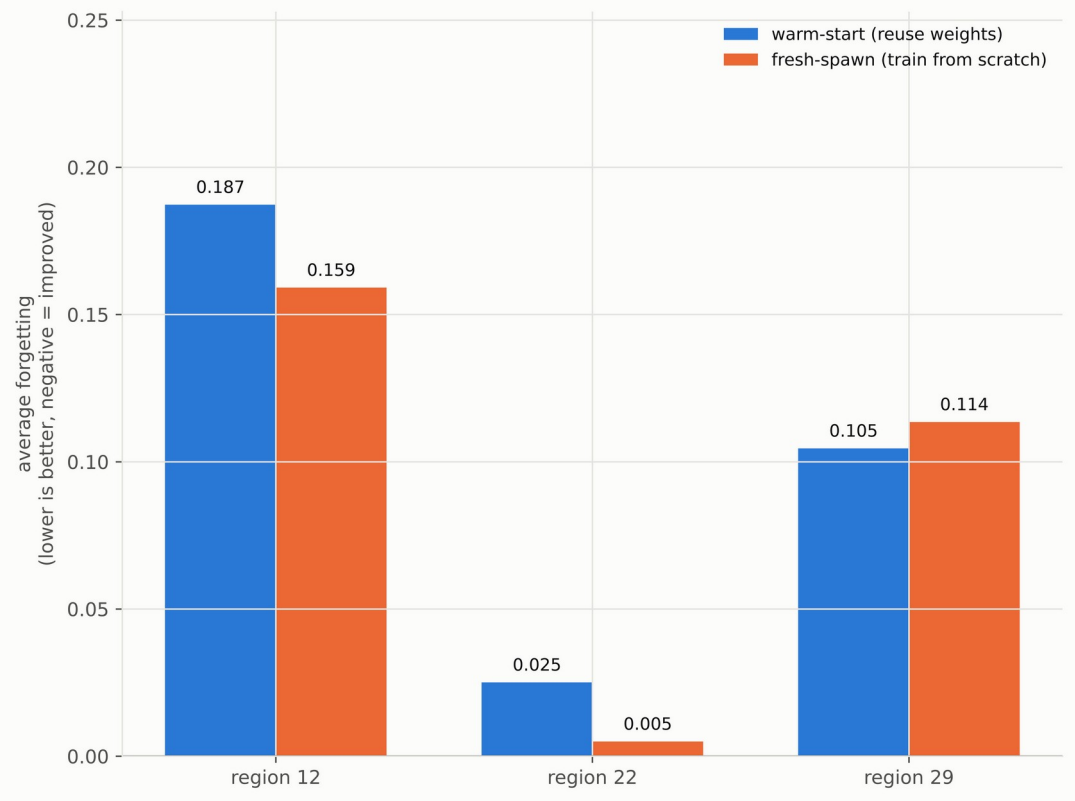
Phase 1 results: expert lifecycle



Reactivation strategy	Region 22	Region 12	Region 29
Top-3 blend beats single best	30/49	42/61	41/61
Mean MSE: single -> blend	0.368 -> 0.352	0.422 -> 0.402	0.384 -> 0.376
Forgetting: single -> blend	-0.010 -> -0.013	0.183 -> 0.170	0.092 -> 0.086

104 of 171 regime changes reactivated an archived expert instead of training from scratch: regimes recur and the archive pays off. Blending the 3 closest archived experts lowers both error and forgetting at every region.

Phase 1 results: warm start vs fresh spawn



- Forgetting = increase in error on a regime seen before, after adapting (lower is better).
- **Warm start** (reuse archived weights) gives lower forecast error in 28/34 events at region 22 and 24/36 at region 29.
- On forgetting it wins only at region 29, so the choice is region-dependent.
- **Default:** reactivate by pooled top-3 blend, the most consistent winner.

Key findings

0 / 250
decoys selected

3 / 536
physically impossible links

62%
links persist at 2x resolution

32 / 50
regions: better 1-day skill

up to 9.9x
faster drift detection (causal channel)

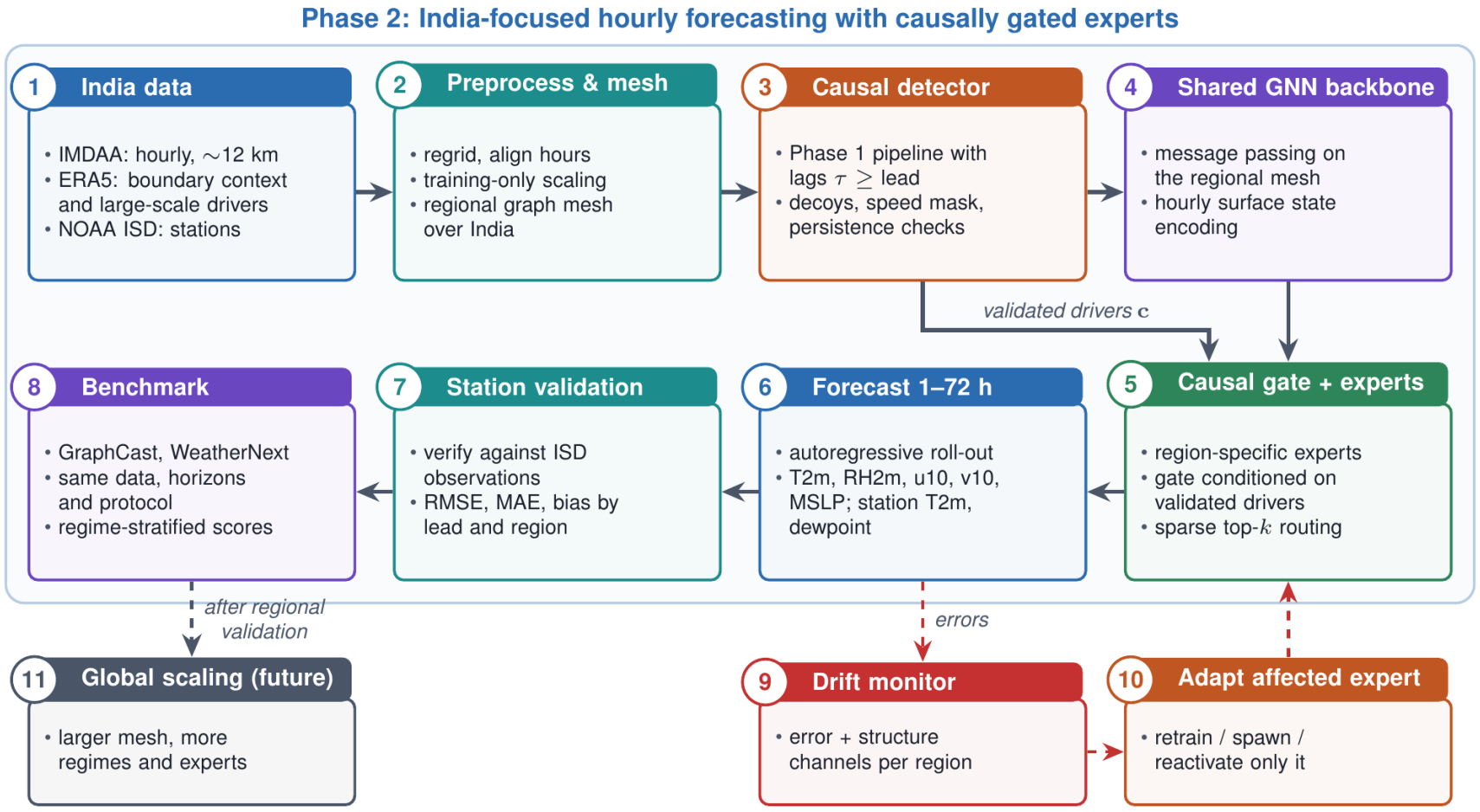
104 / 171
regime changes reuse an archived expert

- The causal graph is trustworthy by every check that can reject it.
- It makes source selection meaningful and helps at the lead its links match (1 day).
- Watching causal structure reveals drift early; recurring regimes make archived experts valuable.

Current limitations

- No independently verified ground truth exists for real climate causal links: every claim rests on indirect checks (decoys, physical plausibility, resolution persistence).
- The validated graph is short-lag (mostly 1–2 days) while the main forecaster runs at 7 days: a time-scale mismatch that limits the prior there. A graph with lags of 7+ days is the fix.
- Forecast comparisons use one training seed and are made against the unguided splitter; random- and correlation-prior controls of equal size are the next runs.
- Drift and lifecycle results cover three regions so far.

Phase 2 plan


Phase 2: India-focused hourly forecasting with causally gated experts
1 India data
• IMDAA: hourly, ∼12 km
• ERA5: boundary context and large-scale drivers
• NOAA ISD: stations
2 Preprocess & mesh
• regrid, align hours
• training-only scaling
• regional graph mesh over India
3 Causal detector
• Phase 1 pipeline with lags $\tau \geq$ lead
• decoys, speed mask, persistence checks
4 Shared GNN backbone
• message passing on the regional mesh
• hourly surface state encoding
validated drivers $\mathbf{c}$
5 Causal gate + experts
• region-specific experts
• gate conditioned on validated drivers
• sparse top-k routing
6 Forecast 1–72 h
• autoregressive roll-out
• T2m, RH2m, u10, v10, MSLP; station T2m, dewpoint
7 Station validation
• verify against ISD observations
• RMSE, MAE, bias by lead and region
8 Benchmark
• GraphCast, WeatherNext
• same data, horizons and protocol
• regime-stratified scores
after regional validation
errors
9 Drift monitor
• error + structure channels per region
10 Adapt affected expert
• retrain / spawn / reactivate only it
11 Global scaling (future)
• larger mesh, more regimes and experts

Phase 2 plan

- **India hourly system:** shared GNN backbone over IMDAA (~12 km, hourly) with ERA5 boundary context; region-specific experts gated by Phase 1-validated causal drivers.
- **Lead-matched causal drivers:** PCMCI tested only at lags ≥ forecast lead, with decoy, speed and persistence checks.
- **72-hour forecasts:** autoregressive roll-out of 2 m temperature, humidity, 10 m wind and MSLP, validated against ISD station observations.
- **Selective adaptation:** per-region drift detection; retrain, spawn or reactivate only the affected expert.
- **Benchmark:** GraphCast and WeatherNext on the same data, horizons and protocol.
- **Global scaling:** larger meshes, more regimes and long-range teleconnections once India is competitive.

14 | References

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